Improving a Decision Tree Model by Changing the Tree Structure Parameters

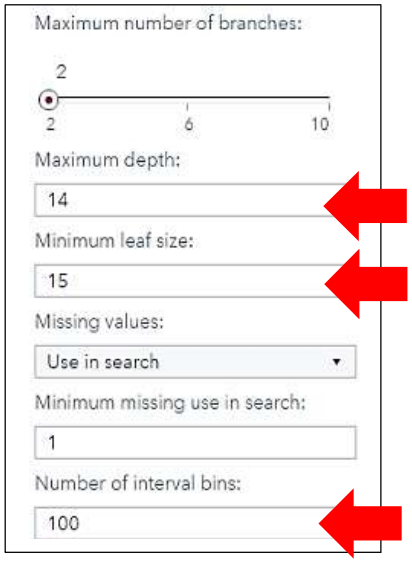
In this demonstration, you change the default settings of the Decision Tree node that was just added in the Lesson 3 pipeline. You modify the tree structure parameters and compare this model performance to the models built earlier in the course.

1. To recall, the previous model, based on the default settings, achieved an average squared error of 0.0694 on the VALIDATE partition.
2. Try to improve the model's performance by modifying some of the settings of the decision tree model. Expand the **Splitting Options** properties in the properties pane of the Decision Tree node.
3. Increase **Maximum depth** from 10 to **14**.

   This specifies 14 as the maximum number of generations in nodes. The original node, generation 0, is called the root node. The children of the root node are the first generation. Possible values range from 1 to 50.
4. Increase **Minimum leaf size** from 5 to **15**.

   This specifies 15 as the smallest number of training observations that a leaf can have.
5. Increase **Number of interval bins** from 50 to **100**.

   This specifies 100 as the number of bins used for interval inputs. Bin size is (maximum value – minimum value) / interval bins. You often have many interval inputs in the form of continuous values. Many machine learning algorithms, including decision trees, operate only in discrete search or variable space. Decision trees typically bin the interval inputs to find the split points.



6. Quantile is the default Interval bin method that bins interval input variables where the number of observations in each bin is approximately equal. Bucket is another method that bins interval input variables into fixed-width bins. Run the **Decision Tree** node.
7. Open the results for the Decision Tree node.

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8. Click the **Assessment** tab and look at model performance using average squared error in the Fit Statistics table. (You might need to expand the Fit Statistics table.)

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0617
churn	VALIDATE	0	0	16,967	0.0663

The average squared error for the tuned decision tree model is 0.0663 on the VALIDATE partition. This fit statistic is slightly better (smaller) compared to that of the first model using the default settings.

9. Close the Results window.

End of Demonstration

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